Course Handout

CHITKARA UNIVERSITY
HIMACHAL PRADESH
NAAC A+ ACCREDITED

Institute/ School Name	School of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Program Name	Bachelor of Engineering (Computer Science & Engineering): B.E (CSE)		
Course Code	24CS015	Course Name	Object Oriented Software Engineering
L-T-P (Per Week)	3-0-2	Course Credits	04
Academic Year	2024-25	Semester/Batch	4th /2023-27
Course Coordinator	Dr. Praveen Kantha		

1. Course Outline:

Introduction to Software Engineering, Layered Technology, Introduction to UML, Requirements Engineering, Introduction to Use-case Diagram, Design Engineering, Introduction to Class diagram, Software Testing, Testing Object Oriented Applications, Introduction to Interaction diagrams, Product Metrics, Project Scheduling, Risk Management, Quality Management.

2. Programme Outcomes (POs):

At the end of the programme, students will be able to:	
PO 1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3. Course Learning Outcomes (CLO):

After completing the course, the students will be able to:

CLO1: Understand the engineering approach and requirement analysis processes in software development.

CLO2: Analyse various software engineering models and apply methods for designing and developing software projects while adhering to professional ethics and utilizing software development skills to enhance employability.

CLO3: Comprehend the benefits of the Unified Modeling Language (UML) as a standard notation for modelling.

CLO4: Predict the cost for an application using process, design, and product metrics.

CLO5: Apply software testing, project scheduling, risk management, and quality assurance skills to validate software products.

4. CLO-PO Mapping Matrix:

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	H						L		L	M		
CLO2		L	L	H				H	M		H	M
CLO3	L		H		L							
CLO4	H	L	M	M							H	M
CLO5	M					M		M	M		M	H

5. ERISE Grid Mapping:

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	2
Research/Innovation	1
Skills	5
Employability	4

6. Recommended Books (Reference Books/Text Books):

B01: Pressman, R. S., Software Engineering: a practitioner's approach (9th Edition). Pressman and Associates.

B02: Schach, S. R., Object-oriented and classical software engineering (6th Edition). McGraw-Hill.

B03: Jacobson, I., Object-oriented software engineering: A use case driven approach (7th Edition). Pearson Education India.

B04: Unni, H., Scrum for startups: Accelerating growth with agile practices (1st Edition). Notion Press.

B05: Mathur, A. P., Foundations of software testing, (2nd Edition). Pearson Education India.

7. Other readings and relevant websites:

Resources	Link of Journals, Magazines, Websites and Research Papers
R1	Link for UMLet, a free and open-source UML tool with a straightforward interface for quickly creating diagrams. http://www.umlet.com
R2	Online NPTEL courses on Software Engineering offered by IIT Bombay, featuring Professors Rushikesh K Joshi, Umesh Bellur, and N.L. Sarda. https://nptel.ac.in/courses/106101061
R3	Online Software Engineering courses offered by Coursera. https://in.coursera.org/courses?query=software%20engineering

R4	Myllärniemi, V., Kujala, S., Raatikainen, M. and SevońnP. (2018). Development as a journey: factors supporting the adoption and use of software frameworks. Journal of Software Engineering Research and Development, 6(1). doi:https://doi.org/10.1186/s40411-018-0050-8.
Resources	**Link of Audio-Video resources**
V1	https://www.youtube.com/watch?v=K9eN_4ob0ZE
V2	https://www.uml.org/resource-hub.htm
V3	https://nptel.ac.in/courses/106105087
V4	https://hp.chitkara.edu.in//listOnlineResources.php

* Resources uploaded on ERP system is accessible to all the students registered for the course.

8. Recommended Tools and Platforms:

- Lucidchart (https://www.lucidchart.com)
- Draw.io (https://app.diagrams.net/)
- Selenium(https://www.selenium.dev/)
- TestLink (testlink.org)
- JUnit (https://junit.org)
- Bugzilla (https://www.bugzilla.org/)

9. Course Plan:

Lecture Number	Topics	Weightage in ETE (%)	Instructional Resources
1-2	Introduction to Software Engineering: The Evolving Role of Software, Changing nature of software	20	B01, V1, R1
3-6	The Software Process: Software Engineering–Layered Technology, Process Models: The Waterfall Model, Evolutionary Process Model: Incremental Models, Spiral Model		B01, B02, V2
7-10	Introduction to UML and modeling software, Introduction to Use-case Diagram and Class diagram Practice Problems: • Draw Use-Case Diagram of Transport Management System • Draw Class Diagram of Transport Management System		Annexure-I
11-13	An Agile View of Process: what is agility, what is an agile process, Agile Process Models: extreme programming (XP), ASD, RAID		B01, B02, R1, B03, BO5, V4
14-15	Scrum: Introduction to Scrum, Scrum Framework, Scrum Roles and Responsibilities, Scrum Artifacts, Tools and Techniques for Scrum		B01, B02, B03, BO5, R2, V3
16-17	Requirements Engineering: Requirements Engineering Tasks: Initiating Requirement, Engineering Process, Eliciting Requirements		B01, B02, R3
18-19	Building Analysis Model: Requirement Analysis, Data modeling Concepts, Flow Oriented Modelling	20	B01, B03, R1, V3
20-22	Design Engineering: Design concepts and model: Cohesion, Coupling, Information hiding, Functional independence, Refinement, Data design, Architectural design		B02, BO5, R2, V1
23-24	Designing class-based components, User interface analysis and design, Interface analysis, and Interface design steps		B02, BO5, R2, V3
25-28	Introduction to Sequence Diagrams, and Activity Diagram Practice Problems: • Draw Sequence Diagram for Transport Management System • Draw Activity Diagram for Transport Management System		Annexure-I

29-30	Software Testing Strategies and Tactics: A strategic approach for Software Testing, Software Testing Strategies: Unit Testing, Integration Testing		B01, B02, BO5, R2, V3
31-34	Practice Problems: • Study of Test Management Tool "TestLink" • Design and Develop Test Cases for Unit Testing and Integration Testing Techniques in an E-Ticketing System	20	Annexure-I
35-37	Validation Testing, System Testing, Test strategies for Object-Oriented Software- Unit Testing in the OO Context, Integration Testing in the OO Context		B01, B02, R2, V1
38-40	White-Box Testing Techniques: Basis Path Testing, Control Structure Testing: condition and loop testing, Black-Box Testing Techniques: Equivalence Partitioning and Boundary Value Analysis		B01, B02, B03, BO5, R2, V3
41-44	Practice Problems: • Study of Test Management Tool "JUnit" • Design and Develop Test Cases for White-Box and Black-Box Testing Techniques in a Recruitment System		Annexure-I
45-46	Testing Object Oriented Applications: Testing OOA and OOD model, Object Oriented Testing Strategies, Object Oriented Testing Methods, Introduction to Automation Testing		B01, B02, B04, BO5, R2, V3
47-50	Study Web testing tool "SELENIUM" and solve Practice Problems: • Cross-Browser Testing for Webpage Compatibility • Cross-Platform Testing on Web Application	20	Annexure-I
51-52	Project Management & Metrics: The management spectrum, Metrics for process & project, Metrics for Software Quality, Estimation		B01, B02, B04, BO5, R2, V1
53-54	Product Metrics: Metrics for the requirement model, Metrics for the design model, Metrics for testing		B01, B02, BO5, R2, V3
55-58	Introduction to State chart diagrams, Data Flow diagram Practice Problems: • Draw State Chart Diagram for Transport Management System • Draw Data Flow Diagram for Transport Management System		Annexure-I
59-60	Software Project Planning: Objective, Software Scope and Resources, Software Project Estimation and Decomposition Techniques (LOC, FP)		B01, B02, BO5, R2, V1
61-62	Empirical Estimation Models: COCOMO Model, Estimation of Object-Oriented Projects	20	B01, B02, BO5, R2, V1
63-64	Project Scheduling: Basic concepts of scheduling, Project Scheduling, Earned Value Analysis		B01, B02, BO5, R2, V2
65-68	Practice Problems: Develop a Gantt chart, and PERT chart for the Examination System		Annexure-I
69-70	Risk Management: Software Risks & Risk Strategies, Risk Identification, Risk Projection, Risk Mitigation, Monitoring and Management (RMMM) plan		B01, B02, B03, B04, BO5, R2, V1
71-72	Practice Problems: Study of Bug Tracking Tool "BUGZILLA"		Annexure-I
73-75	Quality Management: Quality concepts, software quality assurance, software reviews, formal technical reviews, statistical software quality assurance, software reliability, the ISO 9000 quality Standards, Capability Maturity Model (CMM)		B01, B02, BO5, R2, V3

10. Innovative Pedagogies:

- Case study-based learning (Annexure-II)

11. Industry Interventions:

Industry curated course "Software Engineering Fundamentals - Software Development and Testing" at the link below:

https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138429701141135362826 9_shared/overview

12. Action plan for different types of learners

Slow Learners	Average Learners	Advanced Learners
Remedial Classes	Practice Assignment	Advanced Certification from Infosys Springboard (Annexure-III)

13. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment (Offline / Online)
Internal Component 1	Formative Assessments (FAs)	01*	10%	Offline
Internal Component 2	Sessional Tests (STs)	03**	30%	Offline
External Component	End Term Examination	01	60%	Offline
Total		**100%**		

* Formative Assessments (FA) is mandatory.

** Out of 03 STs, the best 02 ST marks will be considered to evaluate final marks.

14. Details of Evaluation Components:

Evaluation Component	Description	Syllabus Covered (%)	Timeline of Examination	Weightage (%)
Internal Component 1	FA1	Up to 50% (Lectures 1-38)	Will be intimated one week prior	10
Internal Component 2	ST 01	Up to 40% (Lectures 1-30)	7th Week	30
	ST 02	41% - 80% (Lectures 31-60)	14th Week	
	ST 03	100% (Lectures 1-75)	17th Week	
External Component	End Term Examination*	100%	As Notified by the Exam Cell	60
Total				100

* Minimum 75% attendance is required to become eligible for appearing in the End Semester Examination

15. Format of Evaluation Components:

Type of Assessment	Total Marks	Certifications	Lab Evaluation	1 Mark MCQ	2 Marks	5 Marks	10 Marks
*Formative Assessments	20	10*	10*	-	-	-	-
Sessional Tests	40	-	-	5	5	3	1
End Term Examination	60	-	-	5	5	5	2

*Formative assessment will carry weightage: 10 marks for Infosys Infyspringboard certifications and 10 marks for lab evaluation.

16. Revision (if any):

Academic Year of Previous Version	2023-2024	Percentage of Revision	10%
Topics Added:			
• RAID, Scrum: Introduction to Scrum, Scrum Framework, Scrum Roles and Responsibilities, Scrum Artifacts, Tools and Techniques for Scrum, Software quality assurance, software reviews, formal technical reviews, statistical software quality assurance, software reliability, Capability Maturity Model (CMM) (Added). • Testing Tools: Selenium, TestLink, JUnit, Bugzilla (Added)			

17. This Document is:

Designation	Name	Signature
Prepared by Course Coordinator	Dr. Praveen Kantha	P. Kantha
Verified by Program Incharge/HoD	Dr. Hakam Singh	for. Hakam
Approved by Pro VC	Prof. (Dr.) Meenu Khurana	Meenu
Date:	January 2, 2025	

Annexure-I

S. No	Experiment
1.	Draw Use-Case Diagram of Transport Management System
2.	Draw Class Diagram of Transport Management System
3.	Draw Sequence Diagrams for Transport Management System
4.	Draw Activity Diagram for Transport Management System
5.	Study of Test Management Tool "TestLink"
6.	Study of Test Management Tool "JUnit"
7.	Design and Develop Test Cases for White-Box and Black-Box Testing Techniques in a Recruitment System
8.	Cross-Browser Testing for Webpage Compatibility using Web testing tool "SELENIUM"
9.	Cross-Platform Testing on Web Application using Web testing tool "SELENIUM"
10.	Draw State Chart Diagram for Transport Management System
11.	Draw Data Flow Diagram for Transport Management System
12.	Develop a Gantt chart, and PERT chart for the Examination System
13.	Study of Bug Tracking Tool "BUGZILLA"

Annexure-II

S. No	Topics
1	Case study-based learning: Software Requirement Specification (SRS) Document

Description:
The students will be introduced to a university-based case study involving departments such as Infrastructure, Examination, Transport, and the Registrar's Office. They will analyze departmental processes, identify automation requirements, and create a detailed Software Requirements Specification (SRS) document to address the identified challenges.

Key Objectives:

- To analyze departmental processes and identify potential areas for automation within university operations.
- To develop a detailed Software Requirements Specification (SRS) document addressing the identified issues.

Group Formation Requirements:

- Each team should consist of a maximum of 10 members.
- Each group must appoint a student coordinator to lead and facilitate discussions and activities.

Annexure-III

The Software Testing Management Certification Course will be offered to advanced learners by Infosys Springboard.

Link:
https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138417511142195202777_shared/overview